# Debt Capacity and the Anatomy of Sovereign Crises

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# Sovereign defaults are rarely complete

- Governments in distress keep paying some obligations and suspend others.
- Missed payments accumulate as arrears that add to the debt.
- Countries often leave a default episode owing more than when it began.

Arellano, Mateos-Planas, and Ríos-Rull (2023): a theory of partial default.

- The sovereign chooses what share of its debt service to withhold.
- It never loses market access. The price reflects the expected default intensity.
- Data: in partial default a third of the time, withholding $\sim$40% of debt service.

# What does partial default teach us about sovereign crises?

Four lessons.

1. Debt capacity. Set by the cost of defaulting, not the value of market access.
2. Entry. Defaults begin with a jump, when shocks overwhelm a precautionary buffer.
3. The episode. A financing regime: borrowing continues until output recovers.
4. Pessimism. Lenders' expectations alone can produce a slow crisis, without a run.

Three versions of the model: deterministic, a Markov chain, and a calibrated diffusion.

# Why continuous time

- The default intensity is a flow decision. Arrears accrue continuously.
- The sovereign never loses access: no stopping problem, no variational inequality.
- One HJB equation and one linear pricing equation, at every state.
- Given the price, the default decision is static: closed-form thresholds.
- Deterministic model: a pair of ODEs, solvable near the steady state.
- Computation: no taste shocks, exact moments, minutes per evaluation.

# Related literature

- **Partial default and arrears:** Arellano, Mateos-Planas, and Ríos-Rull (2023).
- **Binary default:** Eaton–Gersovitz (1981), Aguiar–Gopinath (2006), Arellano (2008).
- **Long-term debt:** Hatchondo–Martinez (2009), Chatterjee–Eyigungor (2012).
- **Dilution:** Hatchondo–Martinez–Sosa-Padilla (2016), Aguiar–Amador (2020).
- **Continuous time:** Bornstein (2020), Tourre (2017), Aguiar et al. (2019).
- **No commitment:** DeMarzo–He–Tourre (2023).
- **Self-fulfilling crises:** Calvo (1988), Cole–Kehoe (2000), Lorenzoni–Werning (2019).
- **Reputation:** Amador–Phelan (2021, 2023).

# The Model

# The sovereign

- Preferences $\mathbb{E}_0 \int_0^\infty e^{-\rho t} u(c_t)\, dt$. Log endowment $z_t$, a Markov process.
- Long-term debt $B_t$: coupon $\lambda$, maturity rate $\delta$, debt service $(\delta + \lambda)B_t$.
- Default intensity $d_t \in [0, 1]$: share withheld. A fraction $\kappa$ becomes arrears.
- Issuance $\ell_t$ at the market price $q_t$. Buybacks allowed.

$$c_t = \phi(d_t, z_t)\, e^{z_t} - (1 - d_t)(\delta + \lambda)B_t + q_t\, \ell_t, \qquad \dot{B}_t = \ell_t - \delta B_t + \kappa(\delta + \lambda)\, d_t B_t$$

- Write $a(d) := \delta - \kappa(\delta + \lambda)d$: absent issuance, debt decays at rate $a(d)$.
- $1 - \kappa$: the haircut lenders suffer on defaulted payments.

# The cost of default

$$\phi(d,z) = (1-\gamma_0 d^{\gamma_1})\,F(z) \;\text{ for } d>0, \qquad \phi(0,z)=1,$$
$$F(z) = 1-\gamma_2\,(e^z - y^*)\,\mathbb{1}\{e^z \geq y^*\}$$

- **Variable cost** $\gamma_0 d^{\gamma_1}$: convex, zero marginal cost at $d=0$.
- **Fixed cost** $(1-F(z))e^z$: paid whenever the sovereign defaults at all.
- Larger in good times. Zero below $y^*$: the *no-fixed-cost region*.
- The jump of $\phi$ at $d=0$ creates a region of no default.

Same structure as Arellano, Mateos-Planas, and Ríos-Rull.

# Lenders and the bond price

- Competitive, risk-neutral lenders with outside option $r$. Coupon $\lambda = r$, so $q^{\text{rf}} = 1$.
- A unit pays $(1-d)(\delta+\lambda)$ in cash and $\kappa(\delta+\lambda)d$ in new units.

$$q(B,z) = \mathbb{E}_{B,z} \int_0^\infty e^{-\int_0^t \xi(d_s)\, ds} (1-d_t)(\lambda+\delta)\, dt, \qquad \xi(d) := r + \delta - \kappa(\delta+\lambda)\, d$$

- Feynman–Kac: $\xi(d)\, q = (1-d)(\lambda+\delta) + \tilde{S}\, q_B + \mathcal{L}_z q$, linear in $q$.
- Spread $= (\delta+\lambda)/q - (\delta+r)$.

# Equilibrium

Given the price $q(B,z)$, the value solves

$$\rho V(B,z) = \max_{c,\, d \in [0,1]} \{u(c) + S(B,z,c,d,q)\, V_B + \mathcal{L}_z V\}$$

- Markov equilibrium: $V$, policies $(c,d)$, and price $q$ that solve both equations.
- A single value function. No exclusion regime, no stopping problem.
- Stationary distribution: the adjoint of the same operator, no simulation.
- First-order condition at an interior drift: $u'(c)\, q = -V_B$.

# Arrears are a source of funds at a fixed price

Let $n_t$ be the total addition to debt, from issuance and arrears. Then

$$c_t = \phi(d_t, z_t)e^{z_t} - (\delta + \lambda)B_t + q_t\, n_t + (1 - \kappa q_t)(\delta + \lambda)B_t\, d_t.$$

- Withholding a unit of debt service raises consumption by $1 - \kappa q_t$, at an output cost.
- Its price is fixed by the recovery on arrears, a haircut that existing creditors bear.
- Its terms do not deteriorate with the sovereign's creditworthiness.
- So arrears are attractive precisely when market prices are low.

Lenders price expected withholding. No transfer: the output cost is a deadweight loss.

# The underwriting standard

- Problem: with no fixed cost, full default makes debt nearly free at the margin.
- Lenders still pay a positive price, so the sovereign borrows to any limit.
- Quantitative models hide this in the top of the debt grid.

**Our fix:** underwriters do not place new issues below a price floor $\underline{q}$,

$$\ell_t \leq 0 \quad \text{whenever} \quad q(B_t, z_t) < \underline{q}.$$

- Outstanding debt still trades. Buybacks and arrears unchanged.
- Makes the level of the borrowing limit irrelevant.
- A price floor on new issues, as in Chatterjee–Eyigungor (2015).

# The default decision is static

For a given drift $S$, consumption is $c = \mathcal{R}(d; B, z, q) + q\,S$, with the *resource function*

$$\mathcal{R}(d; B, z, q) := \phi(d, z)e^{z} - (1 - d)(\delta + \lambda)B + q\,a(d)\,B.$$

**Proposition**

Wherever the standard is slack, every HJB maximizer sets $d^{*} = \arg\max_{d} \mathcal{R}(d; B, z, q)$.

- Independent of the value function and of where debt is going.
- Marginal gain from withholding: $(1 - \kappa q)(\delta + \lambda)B$, payment saved net of arrears.

# Explicit thresholds

Interior intensity and the full-default threshold, at any state and price:

$$d_{\text{int}} = \left[\frac{(1-\kappa q)(\delta+\lambda)B}{\gamma_0\gamma_1 F(z)e^z}\right]^{\frac{1}{\gamma_1-1}}, \qquad B_{\text{fd}}(z,q) = \frac{\gamma_0\gamma_1 F(z)e^z}{(1-\kappa q)(\delta+\lambda)}$$

- No default for $B \leq B_{\text{pd}}(z,q)$, also in closed form. Partial default above it.
- $B_{\text{pd}}$ rises with the price: a lower price makes withholding cheaper.
- No-fixed-cost region: $B_{\text{pd}} = 0$, any positive debt is partially defaulted on.
- Regions that discrete-time models find only numerically.

# A sovereign never defaults a little

- The intensity is a continuous choice, but the fixed cost makes entry lumpy.
- The first default is a jump of a size set by the costs alone,

$$d_{\text{first}}(z) = \left[\frac{1 - F(z)}{(\gamma_1 - 1)\gamma_0 F(z)}\right]^{1/\gamma_1},$$

  about half of the debt service in the calibrated model ($d_{\text{first}} \approx 0.52$, below).
- Independent of debt, the price, and preferences.

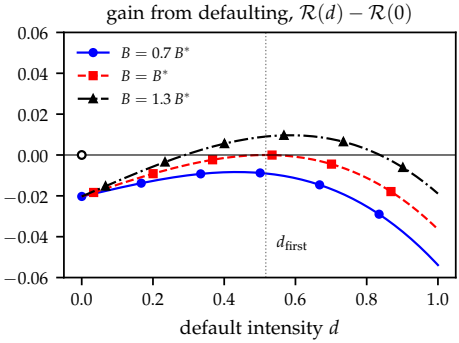
# Why the jump

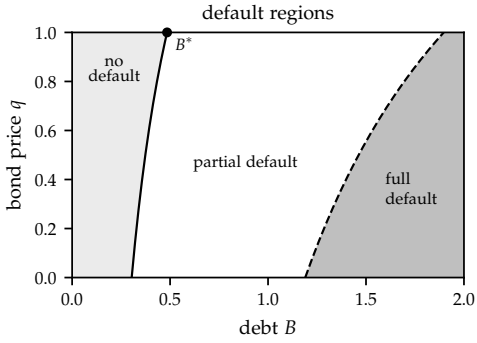
- Without a fixed cost: zero marginal cost at $d = 0$, so a little default at every debt.
- The fixed cost is what makes full repayment possible.
- Once paying the fixed cost, the sovereign withholds until the variable cost bites.
- A steep variable cost ($\gamma_1$ large) makes the jump large.

**Testable in principle**

- Episodes should start with sizable defaults, not small ones.
- Small defaults only where the endowment is near the fixed-cost threshold $y^*$.

# The default regions




- Left: the gain jumps down at $d > 0$, the fixed cost. At $B^*$, a tie.
- Right: the no-default region shrinks as the price falls. Its edge at $q^{\text{rf}}$ is capacity.

# The Deterministic Model

# The deterministic model

With $z$ fixed at 0, the equilibrium is a pair of ordinary differential equations,

$$V'(B) = \frac{\rho V(B) - u(c^*)}{S^*}, \qquad q'(B) = \frac{\xi(d^*)q(B) - (1-d^*)(\lambda+\delta)}{S^*},$$

with $d^* = d^*(B, q(B))$ static, $c^* = (u')^{-1}(-V'/q)$, and $S^* = (c^* - \mathcal{R}^*)/q$.

- Singular where the drift vanishes: the steady states organize the solution.
- No-default steady state: debt capacity. Partial-default steady state: self-fulfilling.
- Illustrations use the calibrated parameters with the endowment at its mean.

# Debt capacity

**Theorem (debt capacity in closed form)**
In the deterministic model the impatient sovereign borrows at the risk-free price up to

$$B^* = B_{\text{pd}}(0, q^{\text{rf}}),$$

the largest debt it is willing to service in full at the risk-free price, and stops there.

- Closed form in the default costs, the recovery on arrears, and debt maturity.
- Independent of patience, risk aversion, and the borrowing limit.

Preferences decide whether to go to the limit. The cost of defaulting decides where it is.

# Why the sovereign stops at $B^*$

- Below $B^*$: no default, price at par, impatience pushes debt up.
- One more unit past $B^*$ starts a default of size $d_{\text{first}}$.
- Lenders would charge for that default at once: the price leaves par.
- The sovereign prefers to stop exactly at the edge of the no-default region.

**Contrast with the binary model**

- There, the limit is where staying in the market stops covering repayment.
- It depends on patience, risk aversion, and the punishment after default.
- It is found only numerically.

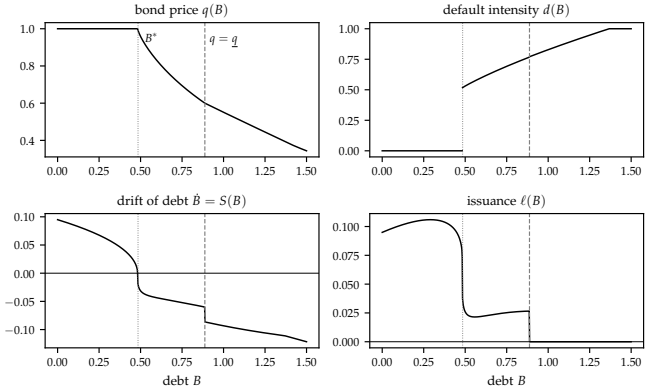
# What moves capacity

- **Default costs** $\gamma_0$, $1 - F$, and **income**: higher, more capacity.
- **Recovery** $\kappa$: more arrears per unit withheld make withholding costlier.
- **Maturity**: shorter maturity, lower capacity.

**The maturity channel**

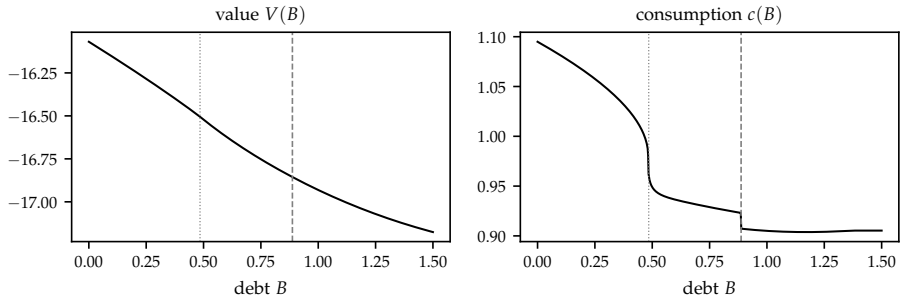
- The gain from withholding scales with the debt-service flow $(\delta + \lambda)B$.
- Raising $\delta$ from 0.12 to 0.22 lowers capacity from 0.584 to 0.358.
- Distinct from dilution and rollover channels (Chatterjee–Eyigungor 2012).

# The deterministic equilibrium




- Par price and positive drift up to $B^*$. Partial default and negative drift beyond it.

# Value and consumption




- $V$ is $C^1$ at $B^*$. Consumption drops steeply but continuously at first default.

# Laws of motion around capacity

**Lemma (square-root law, from below)**

$$S(B^* - x) = (2(\rho - r)\,\psi(\bar{c}))^{1/2}\, x^{1/2}(1 + o(1)), \qquad \psi = u'/|u''|,$$

and $B^*$ is reached in finite time.

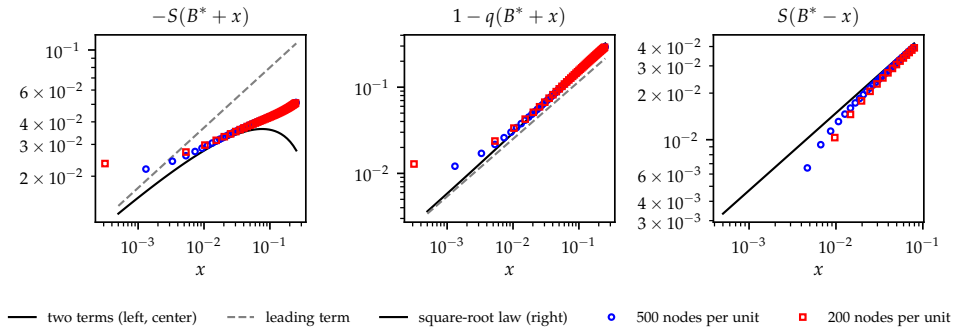
**Theorem (cube-root law, from above)**

With $\Pi_0 = (1-\kappa)(\delta+\lambda)d_{\text{first}}$ and $\beta = (3a(d_{\text{first}})B^*\Pi_0\,\psi(\bar{c}))^{1/3}$,

$$S(B^* + x) = -\beta\, x^{1/3}(1 + o(1)), \qquad q^{\text{rf}} - q(B^* + x) = \tfrac{3\Pi_0}{2\beta}\, x^{2/3}(1 + o(1)).$$

- The value is $C^1$ at $B^*$. Second-order terms in closed form too.

# The cube-root law at work




- Repayment slows near capacity. The price leaves par with a vertical tangent.
- An overhang is priced at once, worked off slowly, without restructuring.

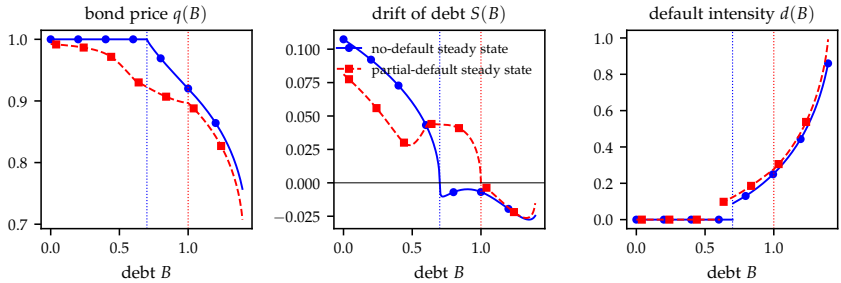
# Pessimism alone can produce a slow crisis

- Lenders expect the sovereign to settle at $B^\circ$, above capacity.
- They expect it to withhold a constant share $d^\circ$ there forever.
- So they pay $q_r(d^\circ) = (1 - d^\circ)(\lambda + \delta)/\xi(d^\circ)$ at $B^\circ$, and less than par below.

**Two consequences validate the expectation**

- At the lower price withholding is cheaper, so default starts earlier.
- The default is priced, so the sovereign borrows past its old limit at the risk-free rate.
- It stops at $B^\circ$, where impatience balances the dilution of its price.

# Two equilibria at the same parameters




- Steady states: no default at $B = 0.70$, partial default at $B = 1.00$ ($d = 28\%$, $q = 0.90$).
- One of a continuum: steady states from 0.93 to 1.02. No run, no loss of access.

# When is pessimism self-fulfilling?

**Proposition (risk-neutral sovereign)**

A self-fulfilling partial default is reached from zero debt only if $\rho - r \geq \Theta^*$: the wedge must cover the output cost of the cheapest default per unit of funds raised. When the default cost is not too convex,

$$\Theta^* = s_{\text{first}},$$

the spread implied by withholding the first default intensity forever. Equilibria exist for a range of wedges above it.

- In the example: $s_{\text{first}} = 47$ bp, attraction up to 80 bp.

# The block test

- At zero debt, issue the steady-state debt at once and collect its market value.
- From then on: pay the risk-free rate on what was raised, the default is priced.
- And lose output $1 - \phi(d, 0)$ to the default, forever.

**Front-loading vs. output cost**

- Gain: the impatience wedge times the funds raised, per unit of time.
- Pessimism is self-fulfilling when the gain covers the output cost.
- At the first default, that cost equals the spread times the value of the debt.

Unlike a rollover crisis, the sovereign can stay out of the default region.

# Comparison with slow-moving and rollover crises

- **Lorenzoni–Werning (2019):** pessimism raises the cost of rollover, debt drifts.
- **Here:** pessimism raises the default intensity, which justifies the price.
- **Calvo (1988):** expected repudiation raises rates, which justify repudiation.
- **Cole–Kehoe (2000):** a run. Here: no run, no loss of market access.
- **Aguiar et al. (2022):** closest, the government keeps issuing at depressed prices.

The difference: no forced rollover. The sovereign must choose to borrow into default.

# The Chain Model

# The chain model

The log endowment takes values $z_j$ and switches from $j$ to $k$ at rate $\eta_{jk}$.

$$\rho V_j(B) = \max_{c,d} \left\{ u(c) + S\, V_j'(B) + \textstyle\sum_{k \neq j} \eta_{jk} \left[ V_k(B) - V_j(B) \right] \right\},$$
$$\xi(d_j)\, q_j(B) = (1 - d_j)(\lambda + \delta) + \tilde{S}_j\, q_j'(B) + \textstyle\sum_{k \neq j} \eta_{jk} \left[ q_k(B) - q_j(B) \right].$$

- The static default decision and the thresholds hold state by state, at $q = q_j(B)$.
- One new element: precaution. It can hold a steady state short of capacity.
- The bridge between the deterministic model and the diffusion.

# Precaution

Endowment on a finite chain, switching from $j$ to $k$ at rate $\eta_{jk}$.

**Proposition (steady states at the risk-free price)**

At an interior steady state $B^\circ$ of state $j$ in a default-free set,

$$(\rho - r)\, u'(c_j) = \sum_{k \neq j} \eta_{jk} \left( u'(c_k) - u'(c_j) \right).$$

The state with the lowest consumption has no such steady state.

- Impatience gain from borrowing = precautionary loss from a worse state.
- The worst state borrows to its edge. Better states can hold a buffer.

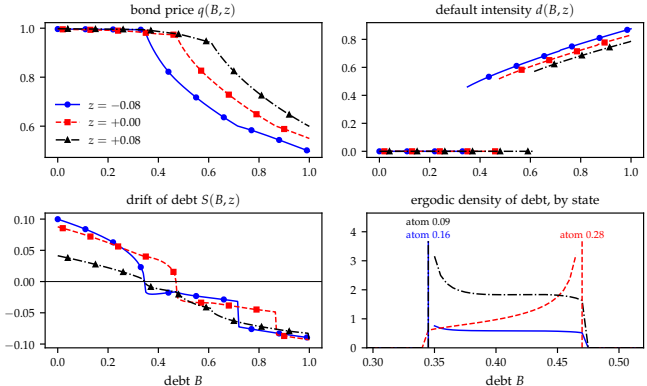
# The price kink and when defaults occur

- Past a worse state's threshold, a switch means immediate default: a price kink.
- In *eight* of *nine* two-state chains, the kink holds the better state below it.

**Default occurs when a shock moves the threshold below the debt**

- Chain: a rare jump. Diffusion: the threshold moves, and the sovereign is caught.
- How often: the tail of shocks against the buffer, not borrowing into default.

# A three-state illustration




- $z \in \{-0.08, 0, 0.08\}$, switches at rate 0.100, rare jumps to the low state at rate 0.03.

# The Diffusion Model

# The diffusion model

Ornstein–Uhlenbeck log endowment, $dz_t = -\mu z_t\,dt + \sigma\,dW_t$, reflected at $\pm\bar{z}$:

$$\rho V = \max_{c,\,d} \left\{ u(c) + S\,V_B - \mu z\,V_z + \tfrac{1}{2}\sigma^2 V_{zz} \right\},$$

$$\xi(d)\,q = (1-d)(\lambda+\delta) + \tilde{S}\,q_B - \mu z\,q_z + \tfrac{1}{2}\sigma^2 q_{zz}.$$

- The continuous-time counterpart of the annual AR(1): $\mu = -\ln \rho_z$.
- Stationary density from the adjoint of the same operator.
- Thresholds move continuously with the endowment.

# Calibration strategy

- Ornstein–Uhlenbeck endowment, $dz = -\mu z\, dt + \sigma\, dW$. CRRA utility, $\nu = 2$.
- Targets, free parameters, and weights of Arellano, Mateos-Planas, and Ríos-Rull.
- Nine free parameters: default costs, discount rate, recovery, maturity, endowment.
- Eleven targets: default frequency and depth, debt, debt service, spreads, output.
- Underwriting standard at $\underline{q} = 0.6$, binding on $<1\%$ of the ergodic mass.

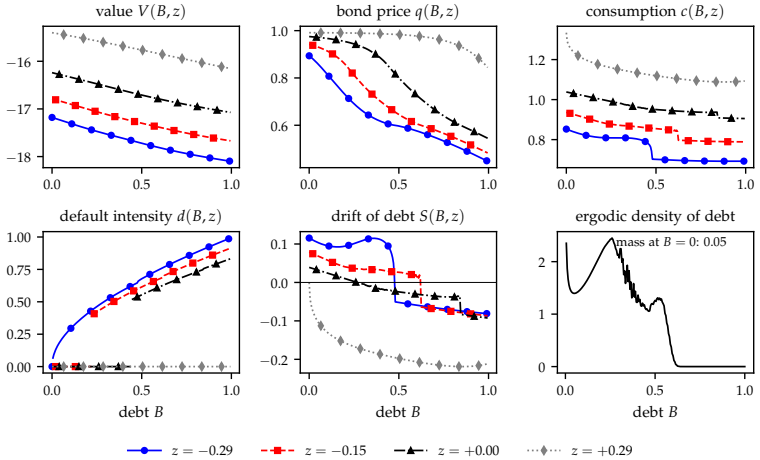
# Parameters

| | | AMR | This paper |
|---|---|---|---|
| *Fixed* | | | |
| $\nu$ | risk aversion | 2 | 2 |
| $r$ | risk-free rate ($\ln 1.04$) | 0.039 | 0.039 |
| $q$ | underwriting standard | – | 0.60 |
| *Calibrated* | | | |
| $\gamma_0$ | variable default cost, level | 0.0476 | 0.0759 |
| $\gamma_1$ | variable default cost, curvature | 2.00 | 3.07 |
| $\gamma_2$ | fixed default cost, slope | 0.120 | 0.082 |
| $y^*$ | fixed default cost, threshold (level) | 0.938 | 0.751 |
| $\rho$ | discount rate (factor $e^{-\rho}$) | 0.047 | 0.062 |
| $\kappa$ | recovery of defaulted coupons | 0.70 | 0.373 |
| $\delta$ | maturity rate (survival $1-\delta$) | 0.12 | 0.153 |
| $\rho_z$ | annual persistence of the endowment | 0.875 | 0.819 |
| $\sigma_\epsilon$ | annual innovation s.d. | 0.052 | 0.066 |

# Fit

| Moment | Weight | Data | AMR | Model |
|---|---|---|---|---|
| *Targeted* | | | | |
| Partial default: frequency | 1 | 0.36 | 0.37 | 0.38 |
| Partial default: mean $\mid d > 0$ | 1 | 0.38 | 0.39 | 0.41 |
| Partial default: s.d. $\mid d > 0$ | 1 | 0.22 | 0.19 | 0.22 |
| Debt to output: mean | 1 | 0.32 | 0.32 | 0.27 |
| Debt to output: s.d. | 1 | 0.18 | 0.25 | 0.18 |
| Debt service to output: mean | 1 | 0.036 | 0.035 | 0.037 |
| Debt service to output: s.d. | 1 | 0.021 | 0.022 | 0.020 |
| Debt due to output: mean | 3 | 0.049 | 0.050 | 0.052 |
| Spread: s.d. | 1.25 | 0.041 | 0.037 | 0.035 |
| Output: persistence | 1 | 0.89 | 0.88 | 0.87 |
| Output: s.d. | 1 | 0.10 | 0.12 | 0.11 |
| Loss | | | 0.228 | 0.077 |
| *Untargeted* | | | | |
| Defaulted coupons to output: mean $\mid d > 0$ | | 0.052 | 0.040 | 0.040 |
| Defaulted coupons to output: s.d. $\mid d > 0$ | | 0.064 | 0.036 | 0.027 |
| Spread: mean | | 0.053 | 0.016 | 0.029 |
| Spread: correlation with output | | $-0.17$ | $-0.38$ | $-0.78$ |
| Spread: correlation with debt | | 0.24 | 0.56 | 0.87 |
| Consumption s.d. relative to output s.d. | | 1.00 | 0.91 | 0.93 |

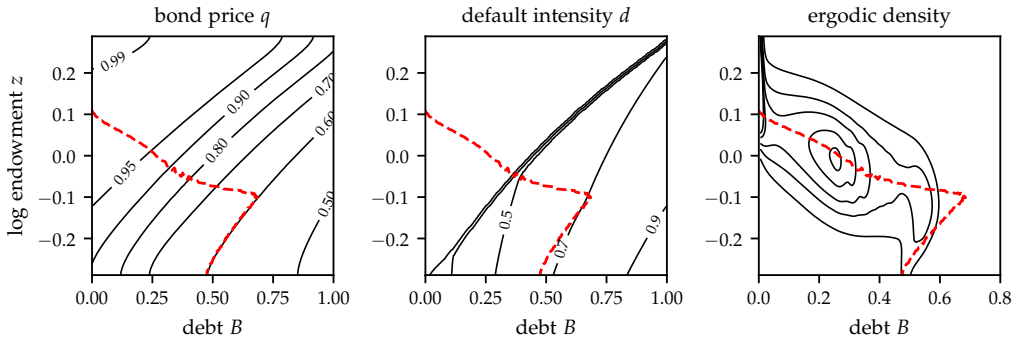
# Values, policies, and prices



# Reading the policies

- The default policy follows the closed-form rule at every endowment.
- Lowest endowment (no fixed cost): default at any positive debt.
- Average and high endowments: no default until $B_{\text{pd}}(z, q)$, then a jump to $d_{\text{first}}$.
- The threshold rises with the endowment. Full default never reached.
- Consumption falls at the threshold, nearly flat in debt beyond it.
- Spreads low most of the time, high in the tail.

# The ergodic set



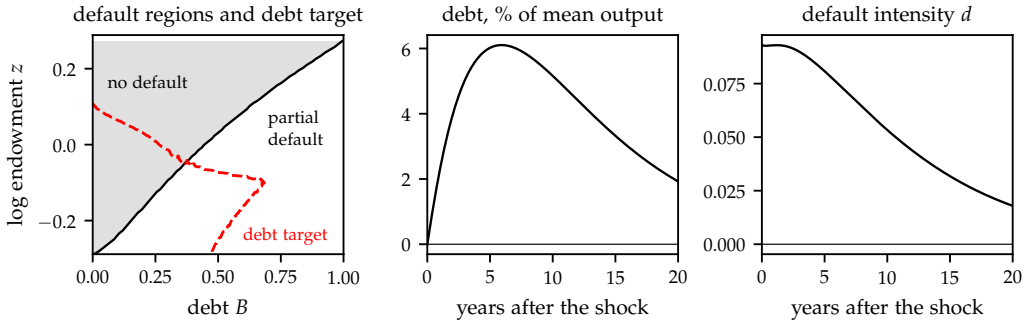

- No-default region: a wedge in $(B, z)$. Mass concentrates along the zero-drift locus.

# The debt target across states

- **Good times:** target below the no-default threshold, held by precaution.
- **Bad times:** threshold below the debt. Partial default, fixed cost sunk.
- Target set by the dilution of the sovereign's own price, inside the default region.
- **Worst states:** target where the price hits the floor and issuance stops.
- **Very good times:** the sovereign repays all the way to zero debt.

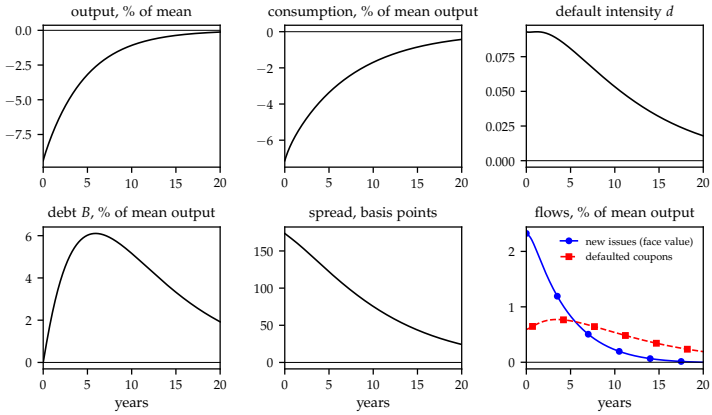
The locus rises from good to bad times, peaks at $B \approx 0.67$, and bends back to 0.48. Default is lumpy at entry and sticky thereafter.

# The anatomy of a crisis




- In bad times the fixed cost is paid regardless and no longer restrains borrowing.
- Debt target set by dilution of the sovereign's own price, higher than in good times.

# After a bad shock




- Shock smoothed by issuing and withholding.
- To the peak: issues add 8.4, arrears 1.6 (% of output).
- Episode ends when output recovers, not when debt is repaid.

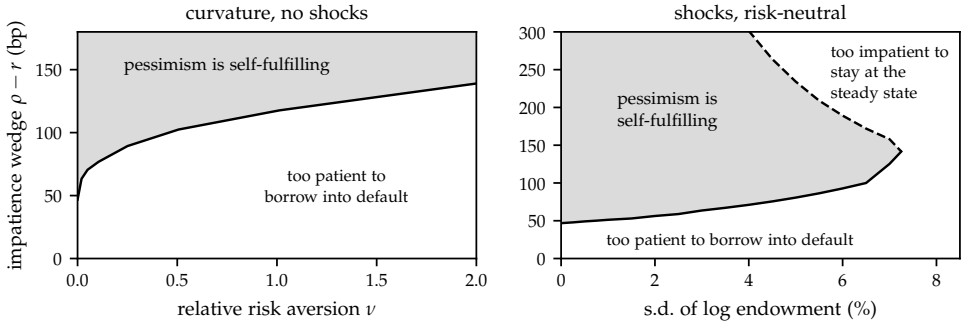
# What the episode explains

- Default jumps on impact and stays elevated for a decade.
- Debt rises for about six years, by some six percent of output.
- The spread rises by about 174 bp on impact and declines with recovery.
- New issues exceed defaulted coupons in face value for five and a half years.

**The facts of partial default episodes**

- Default is protracted. Debt rises during the episode.
- Countries exit owing more than when they entered.
- Market access is the larger cushion against the shock. Arrears are the smaller.

# Curvature and uncertainty raise the bar




- Risk aversion of two roughly triples the impatience needed (139 vs. 47 bp).
- Shocks: window narrows from both sides, closes at endowment s.d. $\approx 7\%$.

# Why curvature and shocks work against it

**Curvature**

- A sovereign that values smooth consumption gains less from front-loading.
- It cannot borrow in a block. It drifts into default gradually.
- The threshold rises with risk aversion and approaches $s_{\text{first}}$ only slowly.

**Uncertainty** (risk-neutral, steady debt independent of the endowment)

- From below: in bad states, a deeper and costlier default at the steady state.
- From above: the steady state must attract debt in every state.
- Waiting at zero debt for a better endowment does not help.

# Self-fulfilling default at the calibrated parameters

- Stationary spread at the first default: 1,286 bp. Spread at the floor: 1,278 bp.
- The standard excludes partial-default steady states.
- Impatience wedge: 225 bp, against a risk-neutral threshold of 1,286 bp.
- Curvature and shocks raise the bar further.

The calibrated sovereign defaults because of shocks, not because of pessimism.

# Conclusion

# Takeaways

- Arrears raise funds at a price fixed by the recovery, a haircut creditors bear.
- Debt capacity is a closed form in the cost of defaulting, not in preferences.
- Crises begin with a jump, when shocks overwhelm a precautionary buffer.
- The episode is a financing regime: borrowing continues until output recovers.
- Pessimism can cause a slow crisis without a run. Curvature, shocks raise the bar.

And a method: continuous time, no smoothing shocks, exact moments.

# Robustness

# Robustness of the calibrated point

| Grid | Debt/y mean | Debt/y s.d. | Spread s.d. | PD freq. | PD mean | Loss | Mass top | Unresolved |
|---|---|---|---|---|---|---|---|---|
| main grid | 0.269 | 0.183 | 0.035 | 0.378 | 0.409 | 0.077 | 0.0000 | 0.015 |
| debt step halved | 0.269 | 0.183 | 0.035 | 0.376 | 0.410 | 0.078 | 0.0000 | 0.007 |
| non-uniform debt grid | 0.269 | 0.183 | 0.035 | 0.375 | 0.410 | 0.078 | 0.0000 | 0.005 |
| barrier at $B = 1.5$ | 0.269 | 0.183 | 0.035 | 0.378 | 0.409 | 0.078 | 0.0000 | 0.015 |

- Finer grid, non-uniform grid, limit at 1.5: moments move within sampling noise.
- The top of the grid carries no mass. The loss is unchanged to the second digit.

# Comparative statics of capacity

| Variation | Steady state $B_{\text{pd}}(0, q^{\text{rf}})$ | Steady state computed | First default $d_{\text{first}}$ | First default computed | $B_{\text{fd}}(0, q^{\text{rf}})$ | Standard binds from | Full default from |
|---|---|---|---|---|---|---|---|
| Calibrated point | 0.485 | 0.475 | 0.517 | 0.519 | 1.90 | 0.890 | – |
| $\gamma_0 \times 0.8$ | 0.451 | 0.440 | 0.556 | 0.557 | 1.52 | 0.785 | 1.11 |
| $\gamma_0 \times 1.2$ | 0.514 | 0.505 | 0.487 | 0.489 | 2.28 | 0.990 | – |
| $\kappa = 0.45$ | 0.553 | 0.540 | 0.517 | 0.518 | 2.16 | 1.005 | – |
| $\kappa = 0.65$ | 0.869 | 0.850 | 0.517 | 0.517 | 3.40 | – | – |
| $\gamma_2 \times 0.5$ | 0.305 | 0.295 | 0.411 | 0.412 | 1.92 | 0.710 | – |
| $\gamma_2 \times 1.5$ | 0.635 | 0.625 | 0.592 | 0.594 | 1.88 | 1.040 | – |
| $\rho = 0.05$ | 0.485 | 0.475 | 0.517 | 0.519 | 1.90 | 0.940 | – |
| $\rho = 0.07$ | 0.485 | 0.475 | 0.517 | 0.519 | 1.90 | 0.860 | – |
| $\delta = 0.12$ | 0.584 | 0.575 | 0.517 | 0.519 | 2.28 | 1.020 | – |
| $\delta = 0.22$ | 0.358 | 0.350 | 0.517 | 0.520 | 1.40 | 0.705 | 1.03 |

- Closed form vs. computed steady state: within a grid step or two in every case.
- Capacity rises with $\gamma_0$, $\kappa$, $\gamma_2$, falls with $\delta$, does not move with $\rho$.

# Other Material

# Computation: the scheme

- Implicit upwind finite differences (Achdou et al. 2022), $201 \times 75$ grid on $(B, z)$.
- Value and price updated jointly each step. Alternating between them is unstable.
- At each node, three candidates: forward, backward, and zero drift.
- For each: closed-form drift, golden-section search over $d$, plus $d \in \{0, 1, d_{\text{int}}\}$.
- Keep the candidate with the largest Hamiltonian: monotone upwinding.
- Handles the fixed cost's non-concavity directly. No taste shocks.

# Computation: accuracy and speed

- Max residuals: HJB $9 \times 10^{-7}$, pricing $10^{-3}$ (locus), $2 \times 10^{-5}$ elsewhere.
- Stationary distribution: the transposed generator, solved directly.
- Impulse responses: the forward equation on the shifted distribution. No sampling.
- One parameter evaluation: about a minute on 25 endowment nodes, four on 75.
- Deterministic model: an exact marching solution of the discrete system, as a check.

# High debt and the underwriting standard

- Without a limit, debt cannot stay above every full-default threshold forever.
- With a limit and no standard, the sovereign borrows up to it, whatever its level.
- With the standard: debt shrinks in full default, the full-default region is transient.
- Any limit above the full-default thresholds is an exact truncation.
- Where it binds, default follows the closed-form rule at the shadow price $-V_B/u'(c)$.
- Unconstrained full-default issuance price $\approx 0.3$, far below the floor of 0.6.

# The price and the value

- Price in $[0, q^{\mathrm{rf}}]$: $q^{\mathrm{rf}}$ only on default-free paths, 0 only in full default forever.
- The price can jump only where debt stops and is not reached from both sides.
- Full default forever, at a zero price: a knife edge, ruled out by no trade at $q = 0$.
- If the price falls with debt, so does the value, with buybacks and the standard.

# Local conditions and the continuum

**Lemma (at a partial-default steady state)**
The price is $q^\circ = q_r(d^\circ)$, and the slope of the price is pinned by impatience,

$$-\frac{q'(B^{\circ+})}{q^\circ} = \frac{\rho - r}{a(d^\circ)\, B^\circ}.$$

- One candidate for every intensity $d^\circ \geq d_{\text{first}}$: a continuum.
- Attracting only if more debt dilutes the price enough, through more default.
- Equilibrium: a two-point problem, borrowing into default from zero debt.
- Lenders' expectations alone select among the candidates.

# Self-fulfilling default with shocks: the construction

Risk-neutral sovereign, diffusion endowment, expected steady debt $B^\circ$ in every state.

- At $B^\circ$: $\xi(d^\circ)q^\circ = (1-d^\circ)(\lambda+\delta) + \mathcal{L}_z q^\circ$, an ODE in $z$.
- Above $B^\circ$ the sovereign is indifferent, $V_B = -q$, and the price solves

$$a(d)B\,q_B = (1-d)(\lambda+\delta) - (\rho + a(d))\,q + \mathcal{L}_z q.$$

- Below $B^\circ$: a block at $q^\circ$. It must beat staying debt-free in every state.
- Checks: attraction, continuation, the floor. At $\sigma = 0$, exactly the deterministic test.